

ITERATION

KEYLA

Starting State

$$m = [-0.8127813177632995, 1.9109011190206628]$$

$$b = [1.0002854206658158, 1.0023556233778974]$$

Compute Prediction

$$\hat{y}_1 = (-0.813 \times 1.911) \times (1 \times 3) + 1.0003 = -3.66$$

$$\hat{y}_2 = (-0.813 \times 1.911) \times (4 \times 10) + 1.0024 = -61.12$$

Compute errors

$$e_1 = -3.66 - 5 = -8.66$$

$$e_2 = -61.12 - 6 = -67.12$$

Compute Gradient

$$dJ/dm_1 = 1.911 \times (3 \times -8.66) + 40 \times -67.12 = \underline{\underline{-5180}}$$

$$dJ/dm_2 = -0.813 \times -2711 = 2203$$

$$dJ/db_1 = e_1 = -8.66$$

$$dJ/db_2 = e_2 = -67.12$$

Update parameters $\alpha = 0.00001$

$$m_1 = -0.812 - 0.00001(-5180) = -0.761$$

$$m_2 = 1.911 - 0.00001(2203) = 1.889$$

$$b_1 = 1.0003 - 0.00001(-8.66) = 1.0004$$

$$b_2 = 1.0024 - 0.00001(-67.12) = 1.0031$$

Final State: $m_4 = [-0.761, 1.889]$

$$b_4 = [1.0004, 1.0031]$$